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DSPL Assignment-1

Q.1 What are the recent trends in Data Science?
→ Data science is rapidly evolving, with several key trends shaping its future:

① Generative AI Expansion:
Generative AI models, such as GPT and BERT, are extending beyond natural language processing into fields like computer vision & drug discovery. Data scientists are focusing on fine-tuning these models for specific applications while addressing ethical concerns

② Edge Computing Integration:
Edge computing reduces latency & bandwidth usage, enabling real-time analytics crucial for applications like autonomous vehicles & smart cities.

③ Synthetic Data Utilization:
To overcome challenges in data availability & privacy, organizations are increasingly generating synthetic data which facilitates model training and testing without compromising sensitive information.

④ Automated Machine Learning:
AutoML platforms are democratizing data science by automating model selection & tuning, allowing those without extensive expertise to build effective models. This accelerates development process & broadens accessibility of machine learning.

⑤ Quantum Computing Applications:
It offers the potential to solve complex problems more efficiently than classical computers. This is useful for fields such as cryptography & complex system simulations

⑥ Data-Centric AI Development: & active
Techniques like data augmentation
learning are being employed to enhance
dataset quality, leading to more robust mode
⑦ Data as a service (DaaS):
DaaS provides data storage, processing &
analytics services over the net work. This
models allows companies to access & share
data insights.
⑧ Privacy-Enhancing Technologies (PETs):
Methods such as differential privacy &
federated learning enable data analysis while
safeguarding individual privacy, helping
organizations comply with data protection
regulations

Q.2 Elaborate in detail, the steps in developing a Machine Learning application with architectural diagram?

→ Architectural Diagram of a Machine Learning Application:

Problem Definition

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Data Collection

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Data Preprocessing

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Model Training & Hyperparameter Tuning

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Model Evaluation

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Model Deployment

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Web/App Integration & API Deployment

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Model Monitoring & Retraining

Steps in developing a Machine Learning application are:

① Problem Definition & Understanding:
It clearly defines the problem statement and business goals, identifies the type of machine learning task (classification, regression, clustering) and determines performance metrics (accuracy, precision, recall, F1-score)

② Data Collection & Preparation:
It gathers raw data from various sourc
(databases, APIs), handles missing values,
duplicate data, normalize or encode
categorical features and divides the data
into training & testing sets

③ Exploratory Data Analysis (EDA):
It performs descriptive statistics to under
data distributions & uses visualization techni
to identify trends & outliers

④ Model Selection & Training:
Chooses an appropriate ML algorithm, trai
the model on training dataset, tunes hyper-
parameters & uses cross-validation to validate
model performance.

⑤ Model Evaluation:
Tests the trained model on the test dataset
and evaluates performance using various metrics

⑥ Model Optimization & Fine-Tuning:
It adjusts hyperparameters to improve accura
& efficiency. It performs feature selection
to remove redundant or irrelevant data.

⑦ Model Deployment:
Converts the trained model into a deployable
format (.pkl, .h5) & deploys the model
using APIs or cloud services

⑧ Monitoring & Maintenance:
It continuously monitor the model's performance
& retrains the model periodically with new
data to avoid performance degradation